GRADUATE STATISTICAL LEARNING | COMPREHENSIVE COMPANION

Classification and Resampling

From linear decision boundaries to honest model validation

PART I Theory, mathematics, worked examples, and mastery exercises

PART II Applied R laboratory: `Hmisc`, `rms`, `MASS`, and `glmnet`

Textbook pathway

The Elements of Statistical Learning, Chapter 4: *Linear Methods for Classification*

An Introduction to Statistical Learning, Chapter 5: *Resampling Methods*

Core question: Can we predict a category, explain why a classifier makes that prediction, and estimate how well it will perform on observations it has never seen?

Original educational synthesis, November 2026. Mathematical examples and R laboratory are independently authored. Not clinical guidance or an externally validated prediction model.

Reading guide and source map

Important chapter identification. The supplied `4.pdf` is ESL Chapter 4, *Linear Methods for Classification*, covering §§4.1–4.5 through optimal separating hyperplanes. The supplied `5(1).pdf` is actually **ISLR Chapter 4, Classification** (not Chapter 5). Because the requested target was ISLR Chapter 5, the resampling chapter is presented from standard ISLR topics and clearly marked as **not contained in the uploaded ISLR file**. The uploaded ISLR Chapter 4 is used as a conceptual bridge, especially logistic regression, LDA/QDA, and classification metrics.

How to study each section. First identify the scientific question; then translate the formula into words; then reproduce the worked calculation on paper; only afterward run the corresponding R lab. Every major method includes assumptions, a failure mode, and a short self-check. Solutions are intentionally separated at the back.

Why combine these chapters? ESL Chapter 4 explains the structure of a classifier: scores, probability models, discriminant functions, and separating planes. ISLR Chapter 5 addresses a different question: how accurately can we assess that classifier using data? A sophisticated model paired with flawed evaluation can be worse science than a simple model that was honestly assessed.

Running biomedical example. In the 189-record `MASS::birthwt` teaching dataset, predict `low`, a binary indicator of low infant birth weight, using maternal characteristics known before delivery (e.g., age, maternal weight, smoking, and previous premature labor). `low` was constructed using infant birth weight `bwt`; therefore `bwt` is prohibited as a predictor. This is a historical small teaching dataset, not a basis for clinical decision making.

Notation. n: observations; p: predictors; K: outcome classes; Y: observed outcome; X: predictors; π_k: class prevalence/prior; p_i: predicted event probability; β: regression coefficients; Σ: covariance matrix; $\hat{}$: an estimate from data.

Contents

Reading guide and source map 2

PART I: Concepts, derivations, and worked problems 5

1 The classification problem: what changed from linear regression? 5

1.1 Continuous outcomes versus categories . . . 5

1.2 Bayes classification and class-conditional information . . . 5

1.3 The difference between score, probability, and boundary . . . 5

2 ESL 4.2: linear regression of a class indicator matrix 6

2.1 Representing a categorical response numerically . . . 6

2.2 Why can this be misleading? . . . 6

3 ESL 4.4: binary logistic regression from the ground up 7

3.1 Derive a probability model from a linear score . . . 7

3.2 Fully worked probability and odds example . . . 7

3.3 Why not ordinary least squares? Why maximum likelihood? . . . 7

3.4 The score and Hessian: understand what software solves . . . 8

3.5 Inference, interpretation, and failure conditions . . . 8

4 ESL 4.4: multinomial logistic regression 8

5 ESL 4.3: linear discriminant analysis (LDA) 9

5.1 Class-specific Gaussian distributions and shared covariance . . . 9

5.2 LDA in one dimension: derive the decision threshold . . . 9

5.3 Where the coefficients come from . . . 9

6 ESL 4.3: QDA and regularized discriminant analysis 10

6.1 From a shared covariance to class-specific covariance . . . 10

6.2 Why more flexible does not mean better . . . 10

6.3 Regularized discriminant analysis (RDA) . . . 10

7 ESL 4.3: Fisher discriminants and reduced rank 11

7.1 Separate class centroids, not simply maximize total variance . . . 11

7.2 Connect the ratio to generalized eigenvectors . . . 11

8 ESL 4.4.5: logistic regression versus LDA 12

9 ESL 4.5: separating hyperplanes, perceptron, and maximum margin 12

9.1 What is a hyperplane? . . . 12

9.2 Rosenblatt's perceptron: correct mistakes with a step . . . 13

9.3 Optimal separating hyperplane and hard-margin support vectors . . . 13

10 ISLR 5: training error is not generalization error 13

10.1 What exactly are we trying to estimate? . . . 13

10.2 A model can succeed at ranking but fail at probability prediction . . . 14

11 ISLR 5.1: the validation-set approach 14

11.1 Why the validation approach can be unstable . . . 14

12 ISLR 5.1: LOOCV and K-fold cross-validation 15

12.1 Leave-one-out cross-validation (LOOCV) . . . 15

12.2 K-fold cross-validation . . . 15

12.3 Stratification and grouping: practical safeguards . . . 16

13 ISLR 5.1: bias–variance and model selection 16

13.1 Why 5-fold and 10-fold are often reasonable . . . 16

13.2 Cross-validation estimates a procedure, not just a fixed model . . . 16

13.3 Nested cross-validation . . . 16

14 ISLR 5.2: the bootstrap method 17

14.1 How bootstrap samples are constructed . . . 17

14.2 Two different uses of bootstrap: uncertainty versus optimism . . . 17

15 Classification performance when outcomes are uncommon **18**
15.1 Confusion matrices and prevalence . . . 18
15.2 Optimal decision cutoffs depend on consequences . . . 18

16 Integrated design: a classification study **18**

17 Part I mastery problems: stop and solve **19**

PART II: Applied R laboratory 20

18 R lab roadmap, assumptions, and reproducibility **20**
18.1 Lab 0: create a clean classification dataset . . . 20
18.2 Lab 1: model binary probabilities with `rms::lrm()` . . . 20
18.3 Lab 2: report odds ratios without confusing them with risks . . . 21
18.4 Lab 3: turn a fitted model into an individual risk prediction . . . 21
18.5 Lab 4: check whether a straight log-odds relationship is too rigid . . . 22
18.6 Lab 5: obtain honest predictions for logistic, LDA, QDA, and lasso . . . 22
18.7 Lab 6: compare discrimination, probabilistic error, and decisions . . . 23
18.8 Lab 7: change the threshold without refitting the model . . . 24
18.9 Lab 8: evaluate calibration on out-of-fold predictions . . . 24
18.10 Lab 9: bootstrap validation of the prespecified `rms` model . . . 24
18.11 Lab 10: bootstrap an odds ratio, not a performance metric . . . 24
18.12 Lab 11: record the software environment . . . 25
18.13 Optional Lab 12: multiclass logistic and Fisher coordinates . . . 25

19 What your finished analysis should report **26**

20 Answers to the Part I mastery checks **26**

21 Answers to the 12 integrated mastery problems **26**

22 References and further checks **27**

PART I CONCEPTS, DERIVATIONS, AND WORKED PROBLEMS

1 The classification problem: what changed from linear regression?

1.1 Continuous outcomes versus categories

In linear regression, the target is a number: for example, the expected infant birth weight measured in grams. In classification, the target is a category: for example, whether the infant is classified as having low birth weight. The same predictors may be useful, but the mathematical output and the scientific claim are different.

For a binary outcome, the most useful intermediate quantity is commonly

$$p(x) = \Pr(Y = 1 \mid X = x), \qquad 0 \le p(x) \le 1. \tag{1}$$

This is the probability of the event for someone with predictors x. Turning $p(x)$ into a class label requires an additional *decision rule*, often $\widehat{Y} = 1$ if $p(x) \ge 0.5$. The cutoff is not a law of statistics; it is a choice whose consequences should be evaluated.

Worked example: risk and action are different. A model predicts $p(x) = 0.28$ for a patient. With threshold 0.50, the classifier says "No event." With threshold 0.20, it says "Event." *Nothing about the fitted probability changed.* Only the decision policy changed. This distinction will matter when we examine sensitivity and specificity.

1.2 Bayes classification and class-conditional information

The general rule under equal costs for incorrect classification is to choose the class with the largest posterior probability:

$$\widehat{G}(x) = \underset{k=1,\dots,K}{\arg\max} \Pr(G = k \mid X = x). \tag{2}$$

Bayes' rule connects the posterior to a class's predictor distribution:

$$\Pr(G = k \mid X = x) = \frac{f_k(x)\,\pi_k}{\sum_{\ell=1}^{K} f_\ell(x)\,\pi_\ell}, \tag{3}$$

where $f_k(x)$ describes how likely predictor pattern x is within class k, and π_k represents the class's prior probability. This decomposition leads directly to LDA and QDA. Logistic regression instead models posterior odds without requiring explicit class-specific predictor densities.

1.3 The difference between score, probability, and boundary

A *score* $s(x)$ orders observations; a *probability* $p(x)$ has a defined numerical interpretation; a *decision boundary* divides the predictor space into regions assigned different labels. A score can be highly discriminative but poorly calibrated as a probability. A boundary can be linear even when the fitted probability changes nonlinearly with x.

Concept map. Linear score ⟶ probability transformation or class comparison ⟶ threshold/boundary ⟶ action. Keep these steps separate in both theory and code.

Mastery check. A predicted event probability is 0.32. The study changes its positive-class threshold from 0.50 to 0.25. What changes: the regression coefficients, the predicted probability, or the assigned class?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: Next, we begin with the apparently simple idea of treating classes as indicator responses. ESL explains why it works in some settings and fails in others.

2 ESL 4.2: linear regression of a class indicator matrix

2.1 Representing a categorical response numerically

Suppose there are K classes. Construct an $n \times K$ indicator matrix Y: for each observation, its row contains a 1 in the column of the observed class and zeroes everywhere else. Fit a least-squares regression to each column, then assign the class with the largest fitted indicator score.

$$\widehat{B} = (X^\top X)^{-1} X^\top Y, \qquad \widehat{G}(x) = \arg\max_k \widehat{f}_k(x). \tag{4}$$

Here X includes the intercept and predictor columns; $\widehat{B}$ contains coefficients for the K score functions; $\widehat{f}_k(x)$ is the fitted score for class k. The inverse expression assumes the relevant columns have full rank; otherwise one needs a generalized inverse or a different fitting strategy.

2.2 Why can this be misleading?

The individual scores need not be valid probabilities: some may be negative or greater than one, even though the fitted scores can sum to one when the intercept and all indicator columns are included. More seriously, with three or more classes, a linear indicator-regression fit can *mask* a class: an intermediate class may never obtain the largest score, even when it is a real class to be predicted. ESL uses a three-class illustration to show this masking phenomenon.

Worked example: a masked middle class. Consider three candidate scores varying with one predictor x: $s_1(x) = 1 - x$, $s_2(x) = 0.45$, $s_3(x) = x$ on $0 \le x \le 1$. Near $x = 0$, class 1 wins. Near $x = 1$, class 3 wins. At $x = 0.5$, scores are $(0.5, 0.45, 0.5)$, so class 2 still loses. In fact $\max(1 - x, x) \ge 0.5$ throughout, so class 2 never wins. These are illustrative scores, not a fitted dataset; they demonstrate the geometry of masking.

Takeaway. Regression on indicators is useful for understanding score geometry, but it is not the default solution to multiclass classification. Probabilistic classifiers should produce valid class probabilities and represent appropriate decision boundaries.

Mastery check. Why does a classifier that produces a score of -0.20 for "event probability" require caution, even if the class predictions appear plausible?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: Logistic regression fixes the probability-range problem by modeling a transformed probability, while retaining a linear predictor.

3 ESL 4.4: binary logistic regression from the ground up

3.1 Derive a probability model from a linear score

Suppose $\eta(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$. We want a probability that always lies between zero and one. The logistic or inverse-logit function is

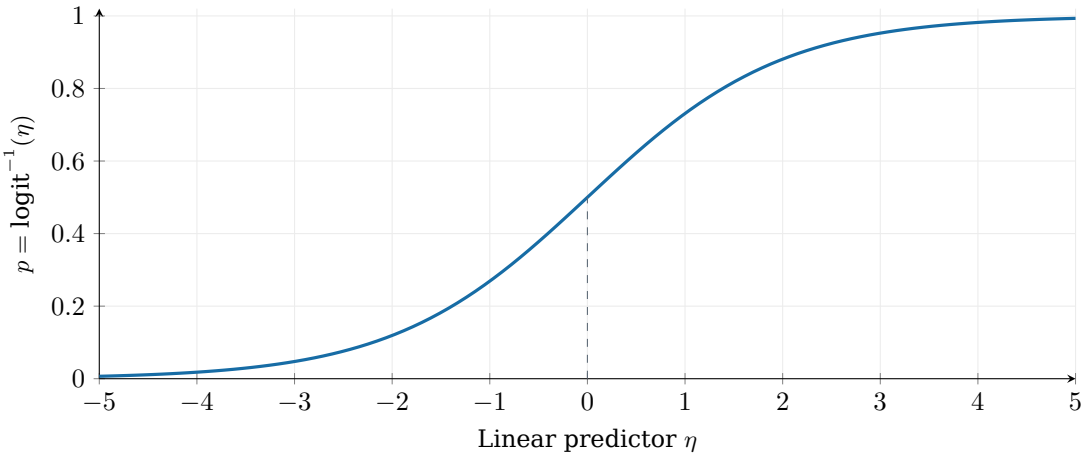
$$p(x) = \frac{e^{\eta(x)}}{1 + e^{\eta(x)}} = \frac{1}{1 + e^{-\eta(x)}}. \tag{5}$$

Algebra gives two equivalent interpretations:

$$\frac{p(x)}{1 - p(x)} = e^{\eta(x)} \qquad \text{odds of the event,} \tag{6}$$

$$\log\left\{\frac{p(x)}{1 - p(x)}\right\} = \eta(x) \qquad \text{log-odds, or logit.} \tag{7}$$

A one-unit increase in predictor x_j, holding the other predictors fixed, increases the *log-odds* by β_j and multiplies the *odds* by e^{β_j}. It does not generally add a constant amount to the probability.



The probability is curved in the linear predictor; the decision boundary at $p = 0.5$ corresponds to the plane $\eta = 0$.

3.2 Fully worked probability and odds example

Consider an *invented* model with a baseline log-odds $\beta_0 = -2$ and a smoking coefficient $\beta_{\text{smoke}} = \log 2 \approx 0.693$. For a nonsmoker, $p_0 = 1/(1 + e^2) = 0.1192$. For a smoker, the linear predictor is $-2 + 0.693 = -1.307$, giving $p_1 = 1/(1 + e^{1.307}) = 0.2130$. The odds ratio is $e^{0.693} = 2$, but the risk difference here is only $0.2130 - 0.1192 = 0.0938$ (9.38 percentage points). At a different baseline risk, the same odds ratio would imply a different risk difference. **Never report an odds ratio of 2 as "twice the probability" without qualification.**

3.3 Why not ordinary least squares? Why maximum likelihood?

For Bernoulli outcomes $y_i \in \{0, 1\}$ with modeled event probabilities p_i, the likelihood is

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}, \tag{8}$$

and its logarithm is

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \tag{9}$$

Why this construction? If the event occurred ($y_i = 1$), that observation contributes p_i to the likelihood; otherwise it contributes $1 - p_i$. The selected coefficients maximize the probability assigned to the observations we actually saw. This is not the same as minimizing squared residuals.

3.4 The score and Hessian: understand what software solves

For the logistic model, with design matrix X, observed event vector y, probability vector p, and diagonal weights $W_{ii} = p_i(1 - p_i)$:

$$\nabla_\beta \ell(\beta) = X^\top (y - p), \tag{10}$$

$$\nabla^2_\beta \ell(\beta) = -X^\top W X. \tag{11}$$

The first expression says: compare actual outcomes with expected probabilities, weighted by predictors. The second describes curvature and information. Newton–Raphson uses both derivatives to update the coefficients. This is the basis for iteratively reweighted least squares (IRLS), which solves a sequence of weighted least-squares approximations. You do *not* need to implement IRLS by hand to fit a scientifically sound model.

3.5 Inference, interpretation, and failure conditions

Under suitable regularity and sampling assumptions, the coefficient estimate can be summarized with an approximate standard error and confidence interval. Small samples, rare outcomes, severe collinearity, data separation, or a highly selected model may make ordinary asymptotic Wald results unreliable. A low p-value for a regression coefficient does not prove causality, and a high p-value does not prove the predictor has no predictive value.

Mastery check. A fitted coefficient for a binary variable is $\log(1.5)$. What is the adjusted odds ratio? Why is a fixed change in predicted probability not implied?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: In binary logistic regression there is one event class and one reference class. Next we extend the same idea to three or more classes.

4 ESL 4.4: multinomial logistic regression

For K classes, choose a reference class K. For each other class k, model

$$\log \frac{\Pr(G = k \mid x)}{\Pr(G = K \mid x)} = \beta_{k0} + x^\top \beta_k = \eta_k(x). \tag{12}$$

The resulting class probabilities have the *softmax* form:

$$\Pr(G = k \mid x) = \frac{e^{\eta_k(x)}}{1 + \sum_{j=1}^{K-1} e^{\eta_j(x)}}, \qquad \Pr(G = K \mid x) = \frac{1}{1 + \sum_{j=1}^{K-1} e^{\eta_j(x)}}. \tag{13}$$

All probabilities are nonnegative and sum to one. Each slope β_{kj} is a conditional log-odds contrast for class k relative to the reference class, not a universal effect on all class probabilities.

Worked three-class example. Suppose the two nonreference linear scores are $\eta_1 = \log 2$ and $\eta_2 = \log 3$. Then $e^{\eta_1} = 2$, $e^{\eta_2} = 3$, and the denominator is $1 + 2 + 3 = 6$. The class probabilities are $(2/6, 3/6, 1/6)$, so class 2 is predicted. Check the probabilities: they sum to exactly one.

Modeling note. Multinomial logistic regression is appropriate for nominal, unordered classes. An ordered response may warrant an ordinal model with different assumptions; do not silently treat ordered categories as nominal or as continuous.

Mastery check. For softmax scores $e^{\eta_1} = 4$ and $e^{\eta_2} = 1$ with class 3 as reference, compute the three class probabilities.
Do the calculation or explain the mechanism before consulting the answers.

Bridge: Logistic regression directly models posterior class probabilities. LDA begins at the other end of Bayes' rule: model the predictor distribution within each class.

5 ESL 4.3: linear discriminant analysis (LDA)

5.1 Class-specific Gaussian distributions and shared covariance

LDA assumes, for each class k,

$$X \mid (G = k) \sim N_p(\mu_k, \Sigma), \tag{14}$$

where μ_k is the class-specific mean vector and Σ is a *common* within-class covariance matrix. The classes may have different means but the covariance structure is shared. Applying Bayes' rule and discarding terms common to all classes yields the discriminant score

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \tfrac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k. \tag{15}$$

Choose the class with the largest $\delta_k(x)$. Why is the boundary linear? When comparing $\delta_k(x)$ with $\delta_\ell(x)$, the differences involve x only in first-degree terms. The shared covariance causes the quadratic terms to cancel.

5.2 LDA in one dimension: derive the decision threshold

Suppose $X \mid G = 0 \sim N(\mu_0, \sigma^2)$ and $X \mid G = 1 \sim N(\mu_1, \sigma^2)$, with $\mu_1 > \mu_0$. Comparing the discriminant scores gives the rule

$$X > \underbrace{\frac{\mu_0 + \mu_1}{2}}_{\text{midpoint}} + \underbrace{\frac{\sigma^2}{\mu_1 - \mu_0} \log \frac{\pi_0}{\pi_1}}_{\text{prior-probability shift}}. \tag{16}$$

If priors are equal, the second term is zero and the boundary is the midpoint of the class means. If class 1 is rarer, the threshold shifts toward its mean, so stronger predictor evidence is needed to classify an observation as class 1.

Worked LDA boundary. Set $\mu_0 = 0$, $\mu_1 = 2$, $\sigma^2 = 1$. With equal priors, classify as 1 when $x > 1$. Now suppose $\pi_1 = 0.2$ and $\pi_0 = 0.8$. Then $\log(\pi_0/\pi_1) = \log 4 = 1.386$. The boundary becomes $1 + (1/2)(1.386) = 1.693$. A data point at $x = 1.4$ changes classification even though the class-conditional Gaussian models have not changed; only the priors changed.

5.3 Where the coefficients come from

Given training labels, estimate $\widehat{\pi}_k = n_k/n$; compute each class-specific mean $\widehat{\mu}_k$; and pool within-class deviations to estimate the shared covariance $\widehat{\Sigma}$. These estimates are then inserted into Eq. (15). If covariance estimation is unstable because p is large or predictors are highly correlated, regularization or dimension reduction may be needed.

Mastery check. In one-dimensional LDA with equal variances, class means 4 and 10, and equal priors, what is the decision threshold? What direction does the threshold move if class 10 becomes rare?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: If the common-covariance assumption is too restrictive, QDA permits a different covariance matrix for each class. That flexibility has a price.

6 ESL 4.3: QDA and regularized discriminant analysis

6.1 From a shared covariance to class-specific covariance

Quadratic discriminant analysis assumes

$$X \mid (G = k) \sim N_p(\mu_k, \Sigma_k). \tag{17}$$

The corresponding score is

$$\delta_k(x) = -\tfrac{1}{2}\log|\Sigma_k| - \tfrac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log \pi_k. \tag{18}$$

Because Σ_k may differ across classes, quadratic terms in x do not cancel. The decision boundary can curve. This may reduce approximation bias when class shapes really differ; with limited data, it can also introduce substantial estimation variance.

6.2 Why more flexible does not mean better

For p predictors, a symmetric covariance matrix has $p(p+1)/2$ distinct entries. LDA estimates one such matrix; QDA estimates one per class. With many predictors or rare classes, QDA rapidly consumes information. In particular, a class-specific covariance matrix may be singular or unstable if there are too few linearly independent observations in a class.

Parameter-count example. With $p = 10$, a covariance matrix has $10(11)/2 = 55$ unique elements. With three classes, QDA estimates $3 \times 55 = 165$ covariance elements; LDA needs 55 pooled covariance elements. These counts exclude class means and priors. The ability to fit class-specific ellipses is not free.

6.3 Regularized discriminant analysis (RDA)

ESL describes shrinkage between the separate and common covariance estimates:

$$\widehat{\Sigma}_k(\alpha) = \alpha\widehat{\Sigma}_k + (1-\alpha)\widehat{\Sigma}, \qquad 0 \le \alpha \le 1. \tag{19}$$

Here $\alpha = 0$ gives the LDA common-covariance estimate and $\alpha = 1$ gives class-specific QDA estimates. Intermediate values shrink noisy class-specific estimates toward the pooled estimate. The shrinkage parameter must be chosen using appropriate training-only procedures, such as inner cross-validation, not by optimizing performance on a final held-out test set.

The key trade-off. LDA may be biased if shared covariance is wrong; QDA may be unstable if each class supplies little information. Regularization trades some flexibility for stability.

Mastery check. If there are $p = 20$ predictors and $K = 4$ classes, how many unique covariance parameters does QDA estimate versus LDA, ignoring other model parameters?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: LDA is not only a probability classifier. The same covariance geometry leads to Fisher's low-dimensional discriminant directions.

7 ESL 4.3: Fisher discriminants and reduced rank

7.1 Separate class centroids, not simply maximize total variance

Principal components analysis seeks axes of large *total* predictor variation; it does not use labels. Fisher's discriminant analysis seeks axes where class centroids differ relative to the noise within classes. Denote by B the between-class scatter matrix and by W the within-class scatter matrix. A direction a is selected by maximizing the Rayleigh quotient

$$J(a) = \frac{a^\top B a}{a^\top W a}. \tag{20}$$

The numerator measures projected variation among the class means; the denominator measures projected within-class variation. A good discriminant direction has strong class separation relative to within-class overlap.

7.2 Connect the ratio to generalized eigenvectors

Scaling a does not change $J(a)$ because the numerator and denominator are both quadratic in a. We can impose $a^\top W a = 1$ and maximize $a^\top B a$ subject to that constraint. The resulting stationary equations are

$$Ba = \lambda W a. \tag{21}$$

This is a generalized eigenvalue problem. The useful discriminant space has dimension no greater than $\min(p, K-1)$ because K class means span at most $K - 1$ independent contrasts about the global mean. Thus, with two classes, the primary Fisher discriminant direction is one-dimensional regardless of the original number of predictors.

Simple numerical demonstration. Suppose $B = \text{diag}(9, 1)$ and $W = \text{diag}(1, 4)$. Along $a = (1, 0)^\top$, $J(a) = 9/1 = 9$. Along $a = (0, 1)^\top$, $J(a) = 1/4 = 0.25$. The first direction provides much greater between-class separation relative to within-class variation. This is *not* the same question as which raw variable has the largest variance.

When to be skeptical. Fisher directions optimize an in-sample criterion based on estimated scatter matrices. Their apparent visual separation is not a generalization estimate; all supervised dimension reduction must be recalculated inside resampling folds.

Mastery check. For three classes and 100 predictors, what is the maximum number of independent Fisher discriminant directions? Why is the answer not 100?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: We can now contrast generative LDA with discriminative logistic regression using a common decision-boundary perspective.

8 ESL 4.4.5: logistic regression versus LDA

	Logistic regression	**LDA**
Modeled object	Directly models $\Pr(Y \mid X)$	Models $f(X \mid Y)$ and class priors
Core assumption	Linear log-odds in specified predictors	Gaussian within-class predictors and shared covariance
Boundary	Linear in included features	Linear under shared covariance
Typical strengths	Probability modeling; interpretable log-odds	Stable with plausible normal structure and suitable sample
Typical limitations	Separation, rare-event instability, misspecified logit	Gaussian/covariance misspecification, unstable covariance

Under LDA's Gaussian/common-covariance assumptions, the posterior log-odds are linear in x, explaining the mathematical relationship with logistic regression. Nevertheless the estimation procedures differ. LDA uses estimates of class densities and priors; logistic regression estimates the conditional likelihood of observed labels. Neither universally dominates the other.

Decision rule versus estimation model. Two models might have nearly the same decision boundary but rather different probability calibration. Similarly, their classification accuracies may look similar despite different likelihoods, assumptions, and coefficient interpretations.

Mastery check. Does a straight decision boundary alone establish that within-class predictors have a multivariate Gaussian distribution? Explain.

Do the calculation or explain the mechanism before consulting the answers.

Bridge: The first part of ESL Chapter 4 culminates in a direct geometric approach: look for a plane separating classes without first constructing a probability model.

9 ESL 4.5: separating hyperplanes, perceptron, and maximum margin

9.1 What is a hyperplane?

A separating hyperplane in p dimensions satisfies

$$\beta_0 + \beta^\top x = 0. \tag{22}$$

In two dimensions it is a line; in three dimensions a plane. An observation is placed on one side or the other according to the sign of $\beta_0 + \beta^\top x$. For nonzero β, the *signed distance* from x to that plane is

$$d(x) = \frac{\beta_0 + \beta^\top x}{\|\beta\|_2}. \tag{23}$$

β is perpendicular to the boundary. Scaling β_0 and β by the same positive constant does not change the boundary or the sign of the predictions, so we need a normalization when defining a unique geometric margin.

Distance example. The line $x_1 + x_2 - 2 = 0$ has $\beta = (1, 1)^\top$ and $\beta_0 = -2$. For $x = (2, 1)$, the score is $2 + 1 - 2 = 1$. Since $\|\beta\| = \sqrt{2}$, the signed distance is $1/\sqrt{2} \approx 0.707$ units. The sign is positive, so this point lies on the positive side.

9.2 Rosenblatt's perceptron: correct mistakes with a step

Let $y_i \in \{-1, +1\}$. When observation i is misclassified, a simple perceptron update is

$$\beta \leftarrow \beta + \eta y_i x_i, \tag{24}$$

$$\beta_0 \leftarrow \beta_0 + \eta y_i, \tag{25}$$

where $\eta > 0$ is the learning rate. The update pushes the score for that training example in the desired direction. If the classes are linearly separable, the perceptron convergence theorem provides convergence under its standard conditions; if not, the algorithm may keep cycling. Even when a separating line exists, its final location may depend on the observation order and starting values.

One perceptron update. Start with $\beta = (0, 0)$, $\beta_0 = 0$ and learning rate $\eta = 1$. A positive observation $y = +1$ at $x = (1, 2)$ is incorrectly or ambiguously classified at zero. The update yields $\beta = (1, 2)$ and $\beta_0 = 1$. Its new score is $1 + (1)(1) + (2)(2) = 6 > 0$, now correctly positive.

9.3 Optimal separating hyperplane and hard-margin support vectors

Among all lines that perfectly separate the training classes, the hard-margin maximum-margin classifier selects one maximizing the distance to the closest observations. One conventional optimization is

$$\underset{\beta_0, \beta}{\text{minimize}} \ \frac{1}{2}\|\beta\|_2^2 \qquad \text{subject to } y_i(\beta_0 + \beta^\top x_i) \geq 1 \ \forall i. \tag{26}$$

The margin between the two supporting planes is $2/\|\beta\|_2$. Observations exactly satisfying $y_i(\beta_0 + \beta^\top x_i) = 1$ are on the margin and are support vectors under the hard-margin solution. The book derives the dual representation and the role of these influential boundary observations.

Scope limit. The supplied ESL Chapter 4 selection develops *separable* optimal hyperplanes. Real biomedical classes typically overlap, so a hard-margin constraint may be impossible. Soft-margin support vector machines and kernels are developed later in ESL; those are extensions, not substantive content of the supplied Chapter 4 selection.

Mastery check. If a hard-margin classifier has $\|\beta\| = 4$, what is the width between its two supporting margin planes?

Do the calculation or explain the mechanism before consulting the answers.

Bridge: A visually impressive boundary can still perform poorly on future patients. That is why the second textbook chapter begins with the meaning of test error.

10 ISLR 5: training error is not generalization error

10.1 What exactly are we trying to estimate?

Let $\mathcal{D}_{\text{train}}$ denote training data and $\widehat{f}_{\mathcal{D}}$ the fitted model. We want the expected prediction loss on a *new independent observation* from the target population:

$$R_{\text{test}} = E_{(X,Y)\sim P_{\text{target}}}\left[L\{Y, \widehat{f}_{\mathcal{D}}(X)\} \mid \mathcal{D}_{\text{train}}\right]. \tag{27}$$

The loss L may be squared error for continuous outcomes, log loss or Brier score for probabilities, or 0–1 error for class labels. If we evaluate the same observations that trained the model, the result is usually optimistically biased, especially when we tuned features or hyperparameters to those observations.

10.2 A model can succeed at ranking but fail at probability prediction

For binary risk prediction, common measures target different questions:

- **AUC/C-index**: Does the model tend to rank an event above a nonevent? It measures discrimination and is invariant to strictly increasing transformations of the scores.
- **Brier score**: $n^{-1}\sum_i (y_i - p_i)^2$. Are predicted probabilities close to observed binary outcomes? Lower is better.
- **Log loss**: $-n^{-1}\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$. Strongly penalizes confident incorrect predictions.
- **Calibration**: Among people receiving a predicted risk near r, does approximately a proportion r experience the event?
- **Sensitivity, specificity, PPV**: How do probability-based decisions perform at a *specified threshold*?

A surprising four-patient example. Outcomes are $(1, 0, 1, 0)$ and predicted probabilities are $(0.8, 0.3, 0.4, 0.1)$. At cutoff 0.5, classifications are $(1, 0, 0, 0)$: 3 of 4 correct, or 75% accuracy. Yet the two cases have probabilities 0.8 and 0.4, both above the controls' probabilities 0.3 and 0.1, so the AUC is 1. The Brier score is $[0.04 + 0.09 + 0.36 + 0.01]/4 = 0.125$. **Ranking, classification, and probability accuracy are different targets.**

Mastery check. Could a model have a high AUC and poorly calibrated probabilities? Give one mechanism.
Do the calculation or explain the mechanism before consulting the answers.

Bridge: ISLR first introduces a simple training/validation split, then shows why resampling is often better use of the available data.

11 ISLR 5.1: the validation-set approach

Split the observations into a training set and a validation set. Fit all model parameters on training data only, predict the validation records, and compute a preselected loss. If a 70/30 split is used with $n = 189$, the validation set contains only about 57 people; for a moderately uncommon outcome, there may be few events available for stable evaluation.

Worked split calculation. Suppose a held-out validation set contains 60 observations: 18 events and 42 nonevents. A classifier misses 5 events and generates 4 false positives. Then TP$= 13$, FN$= 5$, FP$= 4$, TN$= 38$. Sensitivity is $13/18 = 0.722$; specificity is $38/42 = 0.905$; accuracy is $(13 + 38)/60 = 0.85$. An accuracy of 85% alone hides the substantial event-miss rate.

11.1 Why the validation approach can be unstable

The estimate depends on which rows happened to be placed in training and which in validation. A smaller training set may give a worse model than one fitted on all available data. A smaller validation set gives a noisier performance estimate. Repeating random splits can reveal variability, but if we repeatedly inspect the same held-out set while choosing a model, it stops being an untouched validation set.

Biomedical design warning. If individuals have repeated measurements, splits must keep all records for a person together. Multisite generalization may require site-held-out validation. Temporal generalization requires a time-respecting split. Random row splitting can produce leakage and an unrealistic target even when the code is technically correct.

Mastery check. Why is a single 30% validation split often imprecise for a binary outcome with only 15 events in the validation records?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: To use scarce data more efficiently, resampling rotates which records are hidden from the fitting procedure.

12 ISLR 5.1: LOOCV and K-fold cross-validation

12.1 Leave-one-out cross-validation (LOOCV)

With n observations, fit n models. Each model trains on $n-1$ rows and predicts the remaining row. The average loss is

$$\widehat{R}_{\text{LOOCV}} = \frac{1}{n}\sum_{i=1}^{n} L\big(y_i, \widehat{f}_{-i}(x_i)\big), \tag{28}$$

where $\widehat{f}_{-i}$ is the model fitted without observation i. LOOCV uses nearly all observations for each fit, but can be computationally demanding for complex learners and is not guaranteed to have the lowest-variance error estimate.

Connection to the previous linear-regression lecture. For ordinary least squares, a special identity permits LOOCV squared error without refitting n times:

$$\widehat{R}_{\text{LOOCV,OLS}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{1-h_{ii}}\right)^2,$$

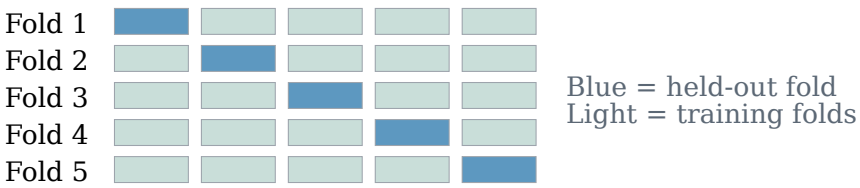
where $e_i = y_i - \widehat{y}_i$ is the ordinary fitted residual and h_{ii} is the diagonal leverage from the hat matrix. If $e_i = 20$ grams and $h_{ii} = 0.20$, the leave-one-out residual is $20/(1-0.20) = 25$ grams and its squared error is 625 grams^2. High-leverage cases can have much larger deleted residuals than ordinary residuals. This exact shortcut is for ordinary least squares, not a generic logistic-regression LOOCV identity.

12.2 K-fold cross-validation

Partition observations into K mutually exclusive folds $I_1, \ldots, I_K$. For each fold k, fit on the other $K-1$ folds and predict those in I_k. A per-observation average loss is

$$\widehat{R}_{K\text{CV}} = \frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k} L\big(y_i, \widehat{f}_{-I_k}(x_i)\big). \tag{29}$$

This formula weights each observation equally, even when folds are not exactly the same size. For $K = 5$, each model trains on approximately 80% of the data; for $K = 10$, approximately 90%. The familiar choices 5 and 10 are pragmatic rather than universally optimal.



12.3 Stratification and grouping: practical safeguards

For binary outcomes, a *stratified* partition approximately preserves the event proportion across folds. This is useful when events are uncommon. But stratifying by outcome does not resolve repeated-measure leakage, related-patient leakage, cluster dependence, or future-time prediction. Those require grouping or appropriate temporal partitioning before or alongside stratification.

Mastery check. If there are 200 independent observations and we perform 5-fold CV, approximately how many observations train each fold-specific model and how many are held out?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: The number of folds influences model-training size, computation, and variability. The statistical trade-off needs interpretation rather than rote memorization.

13 ISLR 5.1: bias–variance and model selection

13.1 Why 5-fold and 10-fold are often reasonable

The loss evaluated in each fold estimates the quality of a model trained on approximately $n(K-1)/K$ observations, not exactly a model trained on all n. This can give a modestly pessimistic estimate relative to the final full-data fit in some settings. LOOCV reduces that training-size difference, but its errors across held-out observations are strongly dependent and its error estimate can have higher variance. Repeated K-fold CV changes partitions to reveal split sensitivity; repeated folds do not create new independent people.

13.2 Cross-validation estimates a procedure, not just a fixed model

Suppose you use CV to choose among logistic regression, LDA, QDA, and 20 sets of engineered predictors. If you report the best observed CV performance from that search as though it were an independent test, it will often be optimistic because the search itself adapted to the same data. The relevant performance target is the *complete learning procedure*: transformations, selection, hyperparameter tuning, model fitting, and prediction.

13.3 Nested cross-validation

In an outer K-fold loop, hold out each test fold untouched. Inside the remaining training observations, choose model settings using *inner* CV. Refit the selected configuration on all outer training data, then predict the outer test fold. Pool the outer held-out predictions to evaluate the selection procedure. The *inner* CV is for tuning, while the *outer* CV is for evaluation.

Worked nested design. Suppose $n = 200$ and outer $K = 5$. Each outer iteration has approximately 160 training and 40 test people. For that iteration, inner 5-fold CV on the 160 people tunes the lasso penalty λ. After selecting λ, the model is fitted on the outer 160 and evaluated once on the 40 untouched outer-test people. Repeat for all 5 outer folds. *Do not fit the scaling parameters, supervised feature selection, or hyperparameter tuner on all 200 in advance.*

Preprocessing belongs within the training fold. Imputation, data-dependent transformations, predictor screening, feature selection, score normalization, and learned model parameters must be fitted using training data only, then applied to held-out rows. R's `glmnet` can standardize within a fitting call; do not

pre-standardize all data using statistics that include the held-out observations.

Mastery check. A researcher uses the entire dataset to select the 20 strongest variables, then runs 5-fold CV on those variables. Why is the reported CV performance likely optimistic?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: Cross-validation refits on data subsets without replacement. The bootstrap uses a different resampling mechanism and answers related but distinct questions.

14 ISLR 5.2: the bootstrap method

14.1 How bootstrap samples are constructed

The nonparametric bootstrap approximates repeated sampling by drawing n observations *with replacement* from the observed n rows. Each bootstrap dataset has n rows, but some original records appear more than once and others appear zero times. A selected original observation has probability $(1 - 1/n)^n \approx e^{-1} = 0.368$ of being omitted from a particular bootstrap sample. Therefore a typical bootstrap draw contains about 63.2% distinct original observations.

Worked bootstrap sample. Original record IDs: $(1, 2, 3, 4, 5)$. One permissible resample is $(1, 1, 3, 4, 4)$. It has five draws but only three distinct people. Records 2 and 5 are *out-of-bag* for this resample. Refit the entire estimation procedure to the five drawn rows and record the statistic of interest. Repeat many times to study variability.

14.2 Two different uses of bootstrap: uncertainty versus optimism

Coefficient uncertainty. If a logistic model's smoking log-odds coefficient is estimated in each bootstrap sample, the empirical distribution across resamples approximates its sampling uncertainty. We can compute a bootstrap standard error or percentile interval. This is *not* a measure of test discrimination.

Prediction optimism. A different bootstrap scheme estimates how much performance is exaggerated when models are evaluated on their training data. Fit to bootstrap samples, evaluate each model on both its resample and a reference set, estimate average optimism, and correct apparent performance. The exact design must include all steps that are being assessed, especially selection or tuning. The `rms::validate()` procedure automates internal bootstrap optimism correction for compatible models.

Bootstrap is not a magic independent test set. The original observations determine the resamples, so resampling measures internal stability under assumptions that the observed sample adequately represents the desired population. It cannot by itself detect temporal or geographic transport failures.

Mastery check. Why does a sample of 189 records drawn with replacement usually contain fewer than 189 distinct individuals? Roughly how many distinct people should be expected?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: We now connect classification losses with clinically meaningful error costs, rare outcomes, and validation boundaries.

15 Classification performance when outcomes are uncommon

15.1 Confusion matrices and prevalence

At any threshold, define TP, TN, FP, and FN. Sensitivity $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$; specificity $= \mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$; positive predictive value (PPV) $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$. PPV depends not only on sensitivity and specificity but also on outcome prevalence in the population to which the decision rule is applied.

Worked prevalence example. In 1,000 people, suppose prevalence is 5% (50 events), sensitivity is 80%, and specificity is 90%. We expect 40 TP, 10 FN, 95 FP (10% of 950), and 855 TN. PPV is $40/(40+95) = 29.6\%$, despite apparently strong sensitivity and specificity. Accuracy is $(40+855)/1000 = 89.5\%$, yet classifying everyone as a nonevent would achieve 95% accuracy with zero sensitivity. **Never evaluate an imbalanced clinical classifier by accuracy alone.**

15.2 Optimal decision cutoffs depend on consequences

If the false-positive cost is C_{FP} and the false-negative cost is C_{FN}, and costs are otherwise symmetric within each error type, choose the positive action when

$$p(x) > \frac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}. \tag{30}$$

This follows by comparing expected cost of classifying positive, $C_{\mathrm{FP}}(1-p)$, with expected cost of classifying negative, $C_{\mathrm{FN}}p$. For $C_{\mathrm{FP}} = 1$, $C_{\mathrm{FN}} = 4$, the threshold is $1/5 = 0.20$. This simplified rule presumes probabilities are relevant to the target population and does not capture all real-world harms or resource constraints.

Mastery check. In the simple cost model above, if a missed event costs 9 units and a false alarm costs 1 unit, what is the optimal threshold under the stated assumptions?
Do the calculation or explain the mechanism before consulting the answers.

Bridge: Before coding, we need one scientific workflow that binds these methods together without confusing model construction with credible model assessment.

16 Integrated design: a classification study

Scientific question: Can maternal characteristics measured before delivery help predict whether the infant will be classified as low birth weight? The exercise is methodological, not a claim that associations are causal or that the model is deployable.

1. **Define** the outcome, target population, index time, predictors available at that time, and appropriate independent sampling unit.
2. **Inspect** event prevalence, missing data, predictor distributions, and impossible values; document cohort selection.
3. **Prespecify** candidate models and evaluation metrics, including calibration and the significance of false negatives.
4. **Fit** an interpretable binary model, examine coefficients on the log-odds scale, and describe uncertainty appropriately.
5. **Check** nonlinear effects and model instability without expanding complexity in response to training data alone.

6. **Evaluate** methods using identically defined outer folds; fit preprocessing and tune penalties only within training records.
7. **Interpret** AUC, Brier, calibration, and threshold-specific sensitivity/specificity as different estimands.
8. **Report** whether estimates are apparent, cross-validated, bootstrap optimism-corrected, or externally validated.

Research-grade distinction. Internal cross-validation estimates performance under the data-generating setting reflected by the available cohort. External validation asks whether the locked model travels to another setting, time, population, or healthcare system. Neither result automatically establishes clinical utility or causal interpretability.

17 Part I mastery problems: stop and solve

Work these without looking at the solutions. You may use a calculator, but show one intermediate step for each numerical question.

Q1. A classifier predicts event probability 0.18. Explain why labeling it "negative" at threshold 0.50 does not mean the model believes the event is impossible.

Q2. A logistic coefficient is $\log 3$. Compute the odds ratio. Explain why this is not necessarily a threefold risk ratio.

Q3. A class-conditional Gaussian model has $\mu_0 = 2$, $\mu_1 = 8$, equal variance, and equal prior probabilities. Find the LDA boundary. What changes when class 1 has a much smaller prior?

Q4. With $p = 6$ predictors and $K = 3$ classes, calculate the covariance-parameter counts for LDA and QDA.

Q5. In Fisher discriminant analysis with $K = 4$ classes, how many useful canonical discriminant directions can there be at most?

Q6. A hard-margin separating plane has coefficient-vector norm 5. Find the full margin width.

Q7. In 5-fold CV with 125 independent people, how many rows are held out in each equally sized fold? How many train each fit?

Q8. A researcher uses CV to select between 50 models and reports the smallest CV loss as an untouched estimate of the winning model's future loss. What is wrong?

Q9. A predictive model has AUC 0.85 but systematically predicts 0.30 risk among people with observed event fraction 0.10. Name the two aspects of performance involved.

Q10. For 500 people, event prevalence is 10%, sensitivity 70%, specificity 90%. Calculate TP, FN, FP, TN, and PPV.

Q11. A bootstrap sample of 200 rows contains 200 draws but perhaps 125 unique records. Is this a bug? Why not?

Q12. State exactly why `bwt` must not be included among predictors when predicting `low` in the `birthwt` dataset.

PART II APPLIED R LABORATORY

18 R lab roadmap, assumptions, and reproducibility

The accompanying, standalone script `Classification_Resampling_Complete_R_Lab.R` contains the code shown below in sequential order. The code is designed around `MASS::birthwt` and the packages `MASS`, `Hmisc`, `rms`, and `glmnet`; an optional multiclass exercise also uses `nnet`. Each lab asks four questions: what are we doing; why; what should we inspect; and what would change our conclusion?

> **Runtime honesty.** The authoring environment did not have an R interpreter, so this laboratory was checked against relevant published R function documentation and reviewed for consistency, but it was **not executed end-to-end**. Package versions and software environments can change. If one command fails, save the exact error and run `traceback()` immediately.

18.1 Lab 0: create a clean classification dataset

Objective. Verify data dimensions, outcome encoding, and factor levels before fitting anything. The outcome `low` has values 0/1, while `low_class` is the matching descriptive factor for LDA and QDA. We keep `bwt` in the original dataset only to recognize the data provenance; we do not put it into any prediction formula.

```r
# LAB 0 - Verify packages and start from the source data
needed <- c("MASS", "Hmisc", "rms", "glmnet")
stopifnot(all(vapply(needed, requireNamespace, logical(1), quietly = TRUE)))
library(rms)  # exposes rcs() for model formulas
set.seed(2026)

birth <- MASS::birthwt
birth$low <- as.integer(birth$low == 1L)
birth$low_class <- factor(birth$low, levels = c(0, 1), labels = c("No", "Yes"))
birth$race <- factor(birth$race, levels = c(1, 2, 3),
                     labels = c("White", "Black", "Other"))
birth$smoke <- factor(birth$smoke, levels = c(0, 1), labels = c("No", "Yes"))
birth$ptl <- factor(birth$ptl > 0, levels = c(FALSE, TRUE), labels = c("No", "Yes"))
# Important: low is derived from birth weight. NEVER include bwt as a predictor of low.
features <- c("low", "low_class", "age", "lwt", "smoke", "ptl")
stopifnot(nrow(birth) == 189L, !anyNA(birth[, features]),
          all(birth$low %in% c(0L, 1L)))
print(table(birth$low_class))
print(with(birth, table(smoke, low_class)))
Hmisc::describe(birth[, features])
```

Validate. Confirm there are 189 observations; the outcome has exactly two classes; the selected variables have no missing values; and the percentage with low birth weight is visible in the table. **Interpret.** A small dataset with a limited number of events cannot support an arbitrarily large model.

18.2 Lab 1: model binary probabilities with `rms::lrm()`

Objective. Fit an interpretable logistic model while retaining the `x` and `y` components needed for internal validation. This is not `rms::ols()`: the outcome is categorical, so we use logistic likelihood rather than squared-error regression.

```r
# LAB 1 - Fit and interpret binary logistic regression with rms
dd_birth <- rms::datadist(birth)
options(datadist = "dd_birth")
form <- low ~ age + lwt + smoke + ptl
fit_rms <- rms::lrm(form, data = birth, x = TRUE, y = TRUE)
print(fit_rms)
print(rms::anova(fit_rms))
print(rms::vif(fit_rms))
plot(rms::Predict(fit_rms, lwt, fun = plogis),
     ylab = "Estimated probability of low birth weight")
```

Validate. The `lrm` printout reports model information, coefficients, uncertainty, and discrimination statistics. `anova` evaluates predictor contributions; `vif` inspects coefficient redundancy; `Predict` shows adjusted estimated event probability as maternal weight changes. The plot depends on correctly setting `options(datadist = "dd_birth")` to refer to the *actual* created object.

18.3 Lab 2: report odds ratios without confusing them with risks

```r
# LAB 2 - Odds ratios and approximate 95% Wald intervals
b <- coef(fit_rms)
se <- sqrt(diag(vcov(fit_rms)))
or_table <- data.frame(
  term = names(b)[-1],
  OR = exp(b[-1]),
  lower_95 = exp(b[-1] - 1.96 * se[-1]),
  upper_95 = exp(b[-1] + 1.96 * se[-1]),
  row.names = NULL
)
print(or_table)
# These Wald intervals rely on large-sample approximations.
```

Validate. For any coefficient $\widehat{\beta}_j$, the adjusted odds ratio is $e^{\widehat{\beta}_j}$. The interval $e^{\widehat{\beta}_j \pm 1.96\, SE(\widehat{\beta}_j)}$ is a simple Wald approximation, not a bootstrap or profile-likelihood interval. The intercept is deliberately removed from the table because its odds interpretation is rarely the primary scientific target.

18.4 Lab 3: turn a fitted model into an individual risk prediction

```r
# LAB 3 - Predict probability for two specified profiles
profiles <- data.frame(
  age = c(27, 27), lwt = c(130, 130),
  smoke = factor(c("No", "Yes"), levels = levels(birth$smoke)),
  ptl = factor(c("No", "No"), levels = levels(birth$ptl))
)
profiles$predicted_risk <- as.numeric(
  predict(fit_rms, newdata = profiles, type = "fitted")
)
print(profiles)
# This is an association-based prediction, not a causal smoking effect.
```

Validate. The output should have two rows with identical age, maternal weight, and premature-labor history, differing only by smoking status. The displayed risk difference is a *model-based counterfactual-style comparison of covariate profiles*; observational data alone do not authorize a causal smoking-effect interpretation.

18.5 Lab 4: check whether a straight log-odds relationship is too rigid

```r
# LAB 4 - Inspect possible nonlinearity using a restricted cubic spline
fit_spline <- rms::lrm(low ~ age + rcs(lwt, 4) + smoke + ptl,
                       data = birth, x = TRUE, y = TRUE)
print(rms::anova(fit_spline))
plot(rms::Predict(fit_spline, lwt, fun = plogis),
     ylab = "Estimated probability of low birth weight")
# With n = 189, avoid choosing complex terms by apparent training fit alone.
```

Interpret. Restricted cubic splines relax the assumption that maternal weight has a constant effect on log-odds at every value. The test of nonlinearity and the shape of the resulting curve deserve attention, but n=189 is small; the point is to *diagnose* a plausible nonlinearity, not to hunt for the best-looking curve. If you intend to compare candidate spline specifications for prediction, the spline fitting and selection must occur inside the outer training folds.

18.6 Lab 5: obtain honest predictions for logistic, LDA, QDA, and lasso

Objective. Construct a single stratified five-fold partition, fit each candidate classifier on the same outer training folds, and keep only held-out predictions. The outer folds are shared to make the comparison fair. Lasso uses inner five-fold CV to select λ using outer training data alone.

```r
# LAB 5 - Honest stratified 5-fold cross-validation, shared folds
K <- 5L
fold <- integer(nrow(birth))
for (g in c(0L, 1L)) {
  idx <- which(birth$low == g)
  fold[idx] <- sample(rep(seq_len(K), length.out = length(idx)))
}
print(table(fold, birth$low_class))

# Each patient's predictions must come from models that did not train on that patient.
p_glm <- p_lda <- p_qda <- p_lasso <- rep(NA_real_, nrow(birth))
# glmnet requires a numeric design matrix; model.matrix encodes factors, not outcomes.
# The feature encoding and candidate predictors are fixed in advance.
X <- stats::model.matrix(~ age + lwt + smoke + ptl, data = birth)[, -1, drop = FALSE]
y <- birth$low

for (k in seq_len(K)) {
  tr <- which(fold != k)
  te <- which(fold == k)

  # Discriminative logistic regression
  mod_glm <- glm(form, data = birth[tr, , drop = FALSE], family = binomial())
  p_glm[te] <- as.numeric(predict(mod_glm, newdata = birth[te, , drop = FALSE],
                                  type = "response"))

  # Generative classifiers, fit on the exact same training records
  mod_lda <- MASS::lda(low_class ~ age + lwt + smoke + ptl,
                       data = birth[tr, , drop = FALSE])
  p_lda[te] <- predict(mod_lda, newdata = birth[te, , drop = FALSE])$posterior[, "Yes"]

  mod_qda <- MASS::qda(low_class ~ age + lwt + smoke + ptl,
                       data = birth[tr, , drop = FALSE])
  p_qda[te] <- predict(mod_qda, newdata = birth[te, , drop = FALSE])$posterior[, "Yes"]
```

```
34
35    # Nested penalty selection: inner CV tunes lambda using outer training rows only
36    set.seed(2026 + k)
37    mod_lasso <- glmnet::cv.glmnet(X[tr, , drop = FALSE], y[tr],
38                                   family = "binomial", alpha = 1,
39                                   nfolds = 5, type.measure = "deviance",
40                                   standardize = TRUE)
41    p_lasso[te] <- as.numeric(predict(mod_lasso,
42                                      newx = X[te, , drop = FALSE],
43                                      s = "lambda.min", type = "response"))
44  }
45
46  stopifnot(all(is.finite(c(p_glm, p_lda, p_qda, p_lasso))))
```

Validate. The fold/class table should show each fold containing both outcome classes. No model is fitted using the records it predicts in the outer test fold. `MASS::lda()` and `MASS::qda()` output posterior class probabilities; `glmnet::cv.glmnet()` is applied to a numeric design matrix because the package API requires numerical inputs. The R script uses `model.matrix()` solely to encode already-prespecified factor levels; it does not hand-implement optimization.

Crucial penalty distinction. `family="binomial"` is correct for logistic classification; `family="gaussian"` is used for continuous-response regression. `alpha=1` means lasso; `alpha=0` means ridge. The lasso `lambda.min` is selected *inside* each outer training fold; choosing it once using all records would compromise the outer assessment.

18.7 Lab 6: compare discrimination, probabilistic error, and decisions

```
1  # LAB 6 - Evaluate ranking, probability quality, and decisions separately
2  score <- function(p, y, threshold = 0.5) {
3    eps <- 1e-12
4    q <- pmin(pmax(p, eps), 1 - eps)
5    pred <- as.integer(p >= threshold)
6    c(AUC = unname(Hmisc::somers2(p, y)["C"]),
7      Brier = mean((p - y)^2),
8      LogLoss = -mean(y * log(q) + (1 - y) * log1p(-q)),
9      Accuracy = mean(pred == y),
10     Sensitivity = mean(pred[y == 1] == 1),
11     Specificity = mean(pred[y == 0] == 0))
12 }
13 comparison <- rbind(
14   Logistic = score(p_glm, y),
15   LDA = score(p_lda, y),
16   QDA = score(p_qda, y),
17   LASSO = score(p_lasso, y)
18 )
19 print(round(comparison, 3))
20 # These are pooled out-of-fold predictions for ONE 5-fold partition.
21 # No confidence intervals or claims of superiority follow from this alone.
```

Validate. AUC is a ranking metric (larger is better); Brier and log loss are probabilistic errors (smaller is better). Accuracy, sensitivity, and specificity here use threshold 0.50. `Hmisc::somers2()` returns a C index compatible with the binary AUC. Use differences cautiously: this is only one partition of a small historical dataset; no confidence interval or superiority claim is established by a few decimal places.

18.8 Lab 7: change the threshold without refitting the model

```r
# LAB 7 - Vary the classification threshold using held-out probabilities
confusion <- function(p, y, threshold) {
  table(Predicted = factor(as.integer(p >= threshold), levels = 0:1),
        Observed = factor(y, levels = 0:1))
}
print(confusion(p_glm, y, threshold = 0.50))
print(confusion(p_glm, y, threshold = 0.20))
print(score(p_glm, y, threshold = 0.20))
# The threshold must be chosen to reflect the decision costs and clinical context.
```

Validate. As the positive threshold drops from 0.50 to 0.20, the number of predicted positives cannot decrease. Sensitivity generally increases or stays the same; specificity generally decreases or stays the same. *The fitted model and its predicted probabilities remain unchanged.* Threshold 0.20 is an instructional scenario, not a clinical prescription.

18.9 Lab 8: evaluate calibration on out-of-fold predictions

```r
# LAB 8 - Calibration using predictions from held-out folds
rms::val.prob(p_glm, y, pl = TRUE, g = 5)
# Interpret this as internal cross-validated calibration, not external validation.
```

Interpret. Calibration is about agreement between predicted risk and event frequency. The diagonal is ideal calibration. With 189 patients, group means and smooth curves can be noisy. `rms::val.prob()` receives the pooled *out-of-fold* logistic probabilities, not the model's apparent fitted values, making this a more credible internal assessment.

18.10 Lab 9: bootstrap validation of the prespecified `rms` model

```r
# LAB 9 - Bootstrap validation of the final prespecified logistic specification
set.seed(2026)
print(rms::validate(fit_rms, method = "boot", B = 200))
set.seed(2026)
cal <- rms::calibrate(fit_rms, method = "boot", B = 200)
plot(cal)
# Optimism correction uses resampling; it does not create an independent cohort.
```

Validate. `validate()` performs model-based bootstrap internal validation and reports optimism-adjusted performance measures; `calibrate()` builds a bootstrap-corrected calibration curve. These are not replacements for an external test population. **Interpret** the apparent results alongside their corrected values, and note the event count and complexity of the model.

18.11 Lab 10: bootstrap an odds ratio, not a performance metric

```r
# LAB 10 - Bootstrap coefficient uncertainty (same dataset, different target)
set.seed(2026)
boot_logor <- replicate(500, {
  ids <- sample.int(nrow(birth), size = nrow(birth), replace = TRUE)
  m <- glm(form, data = birth[ids, , drop = FALSE], family = binomial())
  unname(coef(m)["smokeYes"])
```

```
})
print(quantile(exp(boot_logor), probs = c(.025, .5, .975), na.rm = TRUE))
# This summarizes sample-to-sample variation of a coefficient, not model accuracy.
```

Validate. The code resamples patient records with replacement, refits the prespecified logistic formula, extracts the smoking log-odds coefficient, exponentiates it, and prints the 2.5th, 50th, and 97.5th percentiles. These intervals summarize coefficient variability in the observed sample; they do not estimate a classification AUC or solve confounding. Rare class separation can produce extreme coefficients in resamples; monitor warnings.

18.12 Lab 11: record the software environment

```
# LAB 11 - Reproducibility record
print(sessionInfo())
```

Why. Reproducibility requires the dataset, explicit seed, formulas, fold design, package versions, and a defensible explanation of every model-selection step. A script that runs once without warnings is not yet scientific validation.

18.13 Optional Lab 12: multiclass logistic and Fisher coordinates

Objective. A genuinely multiclass outcome cannot be explored adequately with the binary birth-weight outcome. The classic `iris` teaching dataset supplies three named categories and four quantitative predictors. `nnet::multinom()` fits a softmax model, while `MASS::lda()` estimates discriminant coordinates for the same three-class structure.

```
# OPTIONAL LAB 12 - Multiclass softmax and Fisher's discriminant coordinates
# Separate pedagogical dataset: iris has three classes; not a clinical dataset.
stopifnot(requireNamespace("nnet", quietly = TRUE))
flowers <- datasets::iris
mult <- nnet::multinom(Species ~ Sepal.Length + Sepal.Width +
                         Petal.Length + Petal.Width,
                       data = flowers, trace = FALSE)
print(head(predict(mult, newdata = flowers, type = "probs")))
# The three columns are nonnegative probabilities that sum to one.

fisher <- MASS::lda(Species ~ Sepal.Length + Sepal.Width +
                      Petal.Length + Petal.Width, data = flowers)
coords <- predict(fisher)$x
plot(coords[, 1], coords[, 2], col = as.integer(flowers$Species),
     pch = 16, xlab = "First Fisher discriminant", ylab = "Second Fisher discriminant")
# This is an in-sample visualization; it is NOT evidence of test accuracy.
```

Validate. Each row of the multinomial probability output should have three nonnegative values totaling one (within floating-point precision). The two plotted Fisher coordinates summarize supervised class separation; because the coordinates are computed and drawn on the same dataset, this is *not* an out-of-sample performance estimate. It is a visualization of what the ESL reduced-rank derivation accomplishes.

Mastery. Why is the maximal number of Fisher coordinates in the iris example two, even with four measured predictors?

19 What your finished analysis should report

1. **Target.** Binary low birth weight in the historical `MASS::birthwt` dataset; clear timing and predictor restrictions.
2. **Sample.** Total and event counts, missingness, predictor distributions, and any exclusions.
3. **Models.** Prespecified logistic regression, LDA, QDA, and nested-tuned lasso using identical outer folds.
4. **Estimation.** Adjusted logistic odds ratios with uncertainty, cautiously interpreted as associations.
5. **Discrimination and accuracy.** Out-of-fold AUC, Brier score, and log loss with explicit metric directions.
6. **Clinical operating point.** Confusion matrix, sensitivity, specificity, and PPV for a justifiable threshold.
7. **Calibration.** Internal calibration of held-out probabilities, with instability noted.
8. **Robustness.** Internal bootstrap validation and sensitivity to nonlinearity and modeling choices.
9. **Limits.** Small historical dataset; no external validation; possible confounding, selection and transport limitations.

20 Answers to the Part I mastery checks

Short checks embedded in sections

1. Classification threshold 0.50 to 0.25: only the assigned class changes for $p = 0.32$; the probability and coefficients do not.
2. A score of -0.20 cannot itself be a valid probability; it can only be a score requiring a suitable mapping/interpretation.
3. Logistic coefficient $\log(1.5)$ implies odds ratio 1.5; the probability change depends on baseline probability.
4. Softmax $(4, 1, 1)$ normalized has probabilities $(4/6, 1/6, 1/6)$.
5. LDA means 4 and 10 with equal priors: boundary 7; if class 10 becomes rare, the boundary shifts toward 10.
6. $p = 20$: one covariance has $20(21)/2 = 210$ elements; four QDA matrices have 840.
7. Three classes give at most two independent Fisher directions.
8. A linear decision boundary does not establish Gaussian within-class predictors; many fitting mechanisms generate linear boundaries.
9. $\|\beta\| = 4$ gives margin width $2/4 = 0.5$.
10. AUC can be high despite poor calibration: scores can rank well but consistently overestimate absolute event probabilities.
11. With 200 and 5 folds: 160 train, 40 test per fold (assuming equal fold sizes).
12. Selecting predictors on all observations uses held-out labels to adapt the prediction rule, leaking outcome information into validation.
13. Expected distinct records in a bootstrap of size 189: about $0.632(189) \approx 119$.
14. False alarm cost 1 and missed event cost 9: threshold $1/(1 + 9) = 0.10$ under the simplified cost assumptions.

21 Answers to the 12 integrated mastery problems

A1. A probability of 0.18 means estimated 18% event risk, not zero risk; the label at 0.50 is only a decision rule.

A2. The odds ratio is $e^{\log 3} = 3$. Risk ratio depends on baseline probability and is generally different.

A3. Equal-prior/equal-variance LDA midpoint $(2+8)/2 = 5$. A rarer class 1 shifts the boundary toward its mean 8.

A4. One covariance matrix has $6(7)/2 = 21$ unique entries; QDA has $3(21) = 63$.

A5. At most $\min(p, K-1) = 3$ directions, provided $p \geq 3$ and adequate rank.

A6. Margin width $2/5 = 0.4$.

A7. Exactly 25 held out and 100 training per fold.

A8. Model selection adaptively exploited noise in the same CV estimates; an outer test assessment or nested CV is needed for a fair estimate of the full selection strategy.

A9. High AUC means good discrimination; overpredicting risk 0.30 where observed frequency is 0.10 indicates miscalibration.

A10. 50 events, 450 nonevents. TP$= 35$, FN$= 15$, FP$= 45$, TN$= 405$, PPV$= 35/(35+45) = 43.75\%$.

A11. No bug: resampling uses replacement, so repeated and omitted records are expected.

A12. `low` is computed from infant birth weight `bwt`. Including `bwt` would reveal the outcome definition and produce essentially tautological prediction rather than a prospective model.

22 References and further checks

1. Hastie, Tibshirani, Friedman. *The Elements of Statistical Learning*, Chapter 4, *Linear Methods for Classification*; supplied file `4.pdf`, printed pages 101–134. The supplied excerpt develops indicator regression, LDA (including regularization and reduced rank), logistic regression, and separating hyperplanes.
2. James, Witten, Hastie, Tibshirani. *An Introduction to Statistical Learning*, Chapter 4, *Classification*; supplied file `5(1).pdf`, printed pages 129–189. This is used only as the source-backed transition into logistic regression, LDA/QDA, classification metrics, and comparison methods.
3. James et al. *An Introduction to Statistical Learning*, Chapter 5, *Resampling Methods*. The requested chapter itself was *not attached*; Sections 5.1 and 5.2 are taught from standard textbook structure and original explanations rather than quoted from the uploaded file.
4. https://search.r-project.org/CRAN/refmans/rms/html/Predict.html – `rms::Predict()`.
5. https://rdrr.io/cran/rms/man/validate.lrm.html – `rms::validate()` for logistic models.
6. https://www.rdocumentation.org/packages/rms/versions/8.1-1/topics/val.prob – probability validation and calibration.
7. https://rdrr.io/cran/Hmisc/src/R/somers2.s – binary concordance/C-index.
8. https://rdrr.io/cran/glmnet/man/cv.glmnet.html – penalty tuning, supported binomial losses, and cross-validation interface.

Final synthesis. Classification asks *what probability or category should we predict for this observation?* Resampling asks *how much trust should we place in that prediction procedure for unseen observations?* A scientifically persuasive analysis must answer both.